

THEJUS MAHAJAN PH.D.

Curriculum Vitae

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*30.09.1992 (Talikulam, India) Married Nationality: Indian



PROFILE

- Computational Scientist (PhD, Post-doc) with experience in data pipeline engineering, mathematical modelling, and high-performance computing.
- Core skills: R (tidyverse), Python (JAX, Pandas), Fortran, SQL, Git.
- Internship at HealthTwiSt GmbH: refactored a production R pipeline (DeGIR registry, 143K+ records, ~300 clinics), built an R Shiny dashboard, and created the degirtools R package.
- Available from mid-April 2026** (earlier by arrangement).
- Open to: Data Science, Biostatistics, Bioinformatics, Clinical Data Analysis, Research Software Engineering.

PROFESSIONAL EXPERIENCE

Internship – Clinical Biostatistics & Data Pipeline Engineering HealthTwiSt GmbH

02/2026 - 04/2026 Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,348 lines (–26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 357 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
- Created the `degirtools` R package (9 exported functions, `roxygen2` docs, `devtools::check()` passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using `plotly`, `DT`, and `bslib`. Deployed to `shinyapps.io` with GDPR-safe synthetic data.
- Wrote `Tidymodels` teaching materials (Quarto/GitHub) replacing the legacy `caret` framework for ML workflows in R.

Continuing Education – Bioinformatics and Biostatistics CQ Beratung + Bildung GmbH

08/2025 - 02/2026 Berlin, Germany

Applied bioinformatics and biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Next-Generation Sequencing; Nextflow, Galaxy); structural bioinformatics.
- Biostatistics (R/Bioconductor): ANOVA, PCA, HCA, multivariate analysis, Markov chains.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and `trimAl` trimming with Conda-managed dependencies (GitHub).

Guest Scientist (parallel to continuing education from 08/2025) Helmholtz-Zentrum Hereon, Ecosystem Modelling

05/2025 - 10/2025 Geesthacht, Germany

KEY STRENGTHS

- Data Pipeline Engineering**
Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.5% code reduction. Built `degirtools` R package. Byte-level verification after every change via `identical()`.
- Complex Problem Solving**
Resolved critical data loss by building a Fortran simulation to correct a misaligned particle detector; recalculated results using multi-sensor data.
- Bioinformatics & Biostatistics**
Bridging physics expertise with modern biological workflows. Proficient in NGS data analysis, sequence alignment, and multivariate statistical modelling with R (Bioconductor) and Python.

TECHNICAL SKILLS

Programming
R (tidyverse, dplyr, tidyr, stringr, readr, R Markdown), Python (JAX, Pandas, NumPy), Fortran, C++, SQL, BASH

Data Engineering
Pipeline refactoring, config-driven architecture, `identical()`, R Shiny, `devtools/roxygen2` verification, large datasets (HDF5, NetCDF)

Bioinformatics
NGS analysis, Nextflow (DSL2), sequence analysis, Bioconductor

Biostatistics
Multivariate analysis, PCA, ANOVA, dimensionality reduction, `tidymodels`

Tools
Git/GitHub (branching, collaborative workflows), Linux/Unix, HPC clusters, Conda, LaTeX

LANGUAGES

English ●●●●●
Fluent (C1)

German ●●●●●
B1 (Goethe Certificate), B2 in preparation

French ●●●●●
Basic knowledge

Modelling evolutionary processes in marine ecosystems; simulation of long-term environmental impacts using advanced computational methods. Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Course

📅 02/2025 - 04/2025 📍 Hamburg, Germany

Active job search and German language acquisition (Goethe-Zertifikat B1).

Post-doctoral Scientist

Universität Hamburg, Institute for Marine Ecosystem and Fishery Science

📅 08/2021 - 01/2025 📍 Hamburg, Germany

Marine ecosystem modelling

- Parental leave from 01/2024 to 12/2024.
- Model development: Development of the "Cyanobacteria Life Cycle" (CLC) model within the ERGOM framework for climate warming projections.
- High-performance computing (HPC): Translation of a legacy Fortran model into a high-performance Python engine (Google JAX, TPU/GPU parallelisation). Processing of large NetCDF datasets in Linux HPC environment.

Career Transition

📅 04/2021 - 07/2021 📍 Hamburg, Germany

Relocation from India to Germany and active job search.

Physics Tutor & Chess Coach

Freelance / Tutorwaves Solutions Inc (from 08/2020)

📅 10/2018 - 03/2021 📍 Kerala, India

Freelance physics tutoring and coaching of state-level chess teams; from 08/2020 at Tutorwaves: creation of physics content and mentoring of students in the USA and Australia.

Doctoral Researcher in Astrochemistry

Université Paris-Saclay

📅 10/2015 - 09/2018 📍 Orsay, France

Conducting atomic collision experiments, data analysis, and results modelling.

- Acquisition and processing of terabytes of raw data from tandem particle accelerator experiments.
- C++ code optimisation: improvement of existing algorithms for noise suppression, signal smoothing, and conversion of raw data into multi-dimensional structures.
- Development of a molecular fragmentation model for predicting chemical reactions in space (contribution to the KIDA database).

Malayalam

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Native

Tamil

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Fluent

Hindi

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Basic knowledge

COURSES & TRAINING

High Performance Computing (HPC) Training

Jülich Supercomputing Center (JSC),
Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology – The Secret of Life

Comprehensive coursework in genetics, biochemistry, and molecular biology.

PUBLICATIONS

T. Mahajan et al. *J. Phys.: Conf. Ser.*, 1412:142026, 2020.

T. Idbarkach et al. *Astron. Astrophys.*, 628:A75, 2019.

T. Mahajan et al. *J. Phys. B*, 2019.

T. Idbarkach et al. *J. Phys. B*, 51(24):245201, 2018.

T. Idbarkach et al. *J. Phys.: Conf. Ser.*, 1412(11):112028, 2020.

📄 Google Scholar Profile

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO)

📅 10/2015 - 09/2018 📍 Orsay, France

Supervisor: Dr. Karine Béroff. Thesis: *Excitation and fragmentation of C_nN^+ ($n=1-3$) in collisions with He atoms at intermediate velocity.*

M.Sc. in Physics

National Institute of Technology Calicut, India

📅 07/2012 - 12/2014 📍 Calicut, India

First Class with Distinction (CGPA 8.71/10).

Research Project – Theoretical Atomic Physics

IIT Mandi (under Dr. Hari Varma)

📅 02/2015 - 08/2015 📍 Mandi, India

Theoretical atomic physics; partly remote.

B.Sc. in Physics

University of Calicut, India

📅 06/2009 - 04/2012 📍 Thrissur, India

Grade: B+ (CGPA 3.49/4.0).

Hamburg, 11 April 2026

Dr. Thejus Mahajan